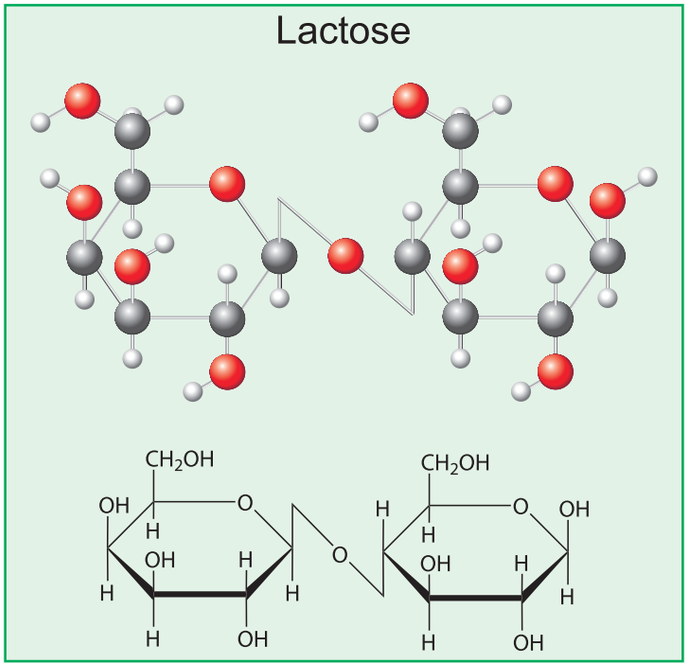
Structure of Lactose:



β-D-glucopyranosyl-(1→4) β-D-glucopyranose

5.6.3 Sucrose:

Properties of sucrose:

a) When heated to 200°C it loses water to form a brown mass called caramel. On strong heating, it gives pure carbon with a burnt smell.

b) Concentrated sulphuric acid dehydrates sucrose into carbon. This on further oxidation by H_2SO_4 gives CO_2

$$C_{12}H_{22}O_{11} + H_2SO_4 \rightarrow C + H_2O + SO_2$$

c) When boiled with HCl, sucrose gives laevulic acid.

d) Concentrated nitric acid oxidises cane sugar (sucrose) to oxalic acid.

$$C_{12}H_{22}O_{11} + 9O_2 \rightarrow 6(COOH)_2 + 5H_2O$$

e) Sucrose is fermented by invertase into glucose and fructose which are converted to ethanol by zymase. Both these enzymes are available in yeast.

f) Sucrose on acetylation gives octa-acetyl derivative.

g) Sucrose on methylation gives octa-o-methyl derivative.

h) Sucrose does not react with HCN, NH_2OH, phenyl hydrazine, Tollen's reagent and Fehling's solution.

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